RESEARCH EVIDENCE / COMPLETION REPORT

GEMSTAR TASK 25 Complete Research Report

Reliability-aware Camera, LiDAR and Radar fusion: sensor health, controlled faults, association, EKF consistency, BEV, fallback and independent computational verification.

Dataset	C02 clean V2, selected EKF and full-rate sensor-health records
Research scope	Tasks 25.1 through 25.10R, including publication checks
Archive	143 files; ZIP SHA-256 verified
Repository branch	research/task25-fusion-reliability
Publication commit	95dac5a5eb50f85861a6b6963d1ea85859cc7e46
Scientific authorization	UNVERIFIED; no measured hardware energy savings

Interpretation: Task 25 is complete as a documented, archived and published research stage. Its perception quality and timing gates are not validated, and its outputs are not approved as a safety-certified or energy-optimized deployment.

Source basis: Nine Task 25 final-stage research reports retrieved from the published GitHub branch, together with the recorded completion, SHA-256, Git and publication-audit outputs. All visualizations in this document are reconstructed from reported numbers and explicitly labelled. The original MATLAB plot files remain in the repository.

Executive summary

Task 25 transformed the previously developed C02 V2 three-sensor EKF pipeline into an auditable reliability research package. It did not introduce an ARM-FPGA hardware measurement. The study first established the behavior and freshness of recorded sensor payloads, applied controlled offline faults to matched replay observations, examined provisional tracking and sensor associations, audited uncertainty against ground truth, constructed selected-frame BEV geometry, and implemented a conservative policy for sensor degradation and hold decisions.

The latter stages separated policy candidate classifications from independently authorized perception results, specified six non-negotiable quality/latency validation gates, established Python-MATLAB numerical reproducibility, and archived and published the evidence on a dedicated Git branch.

Primary evidence	Verified result
Full-rate SensorView recording	2,122 frames
Fault-replay evaluation	7 scenarios × 24 replay observations = 168
GT-center comparisons	23 per scenario = 161 related entries
Radar accepted finite NIS	4 observations
Controller outputs	143 normal / 16 degraded / 9 hold
Numerical cross-language audit	18 / 18 metrics agreed
File integrity and publication	143 GitHub files; verified remote commit

Methodology at a glance

Sequence	Workflow
1. Establish evidence	Identify clean C02 V2 recording, preserve configuration and replay baselines
2. Study failure conditions	Freshness, payload repetition, dropout and injected biases
3. Evaluate perception	Association geometry, EKF residuals, accepted NIS, selected-frame BEV
4. Implement policy	Conservative LiDAR-dependent fallback and internal decision audit
5. Set hard gates	Define evidence needs and prevent quality authorization without proof
6. Reproduce & publish	Independent Python/MATLAB calculation, hash audit, ZIP, GitHub branch

Roadmap / table of contents

25.1 Baseline replay and injection | 25.2 Sensor health | 25.3 Controlled faults | 25.4 Association | 25.5 EKF uncertainty | 25.6 BEV geometry | 25.7 Fallback | 25.8 Hard quality gates (A-F) | 25.9 Cross-language checks (A-E) | 25.10 Archive and GitHub publication (A-R) | Limitations and forward plan.

Task 25.1 | Baseline replay and controlled fault injection

Purpose and steps

• Identify the C02 clean V2 selected-frame recording and establish the unchanged baseline replay.

• Prepare a consistent seven-scenario matrix that reuses the selected replay sample and frame identifiers.

• Generate 24 replay observations per scenario; log sensor usage, accepted/rejected updates and tracking coordinates.

• Produce the replay sample CSV (168 rows) and scenario summary CSV (7 rows).

• Verify each scenario has exactly 24 entries; later check both source hashes against the reproducibility manifest.

• Write the retrospective 25.1 final report without altering any original replay data.

Seven controlled scenarios

Scenario	Fault or condition
baseline	Original selected-frame replay
radar_dropout	Suppress Radar measurement updates
camera_dropout	Suppress Camera updates
lidar_every4_dropout	Suppress periodic LiDAR observations
radar_bias_6mps	Inject +6 m/s Radar bias; monitor NIS rejection
camera_bias_0p20rad	Inject 0.20 rad Camera-bearing bias
joint_dropout	Combined controlled dropout condition

Measured result: 168 replay rows, 7 summaries, 23 GT-center error evaluations per scenario (161 related scenario-evaluation entries). These are not 161 independent recordings.

Evidence: task25_replay_samples.csv; task25_fault_summary.csv; Task25_1_Baseline_Replay_And_Fault_Injection_Final_Report.md

Task 25.2 | Full-rate sensor health and freshness

Purpose and steps

• Extract full-rate Camera, LiDAR and Radar payload changes from the 2,122 SensorView frames.

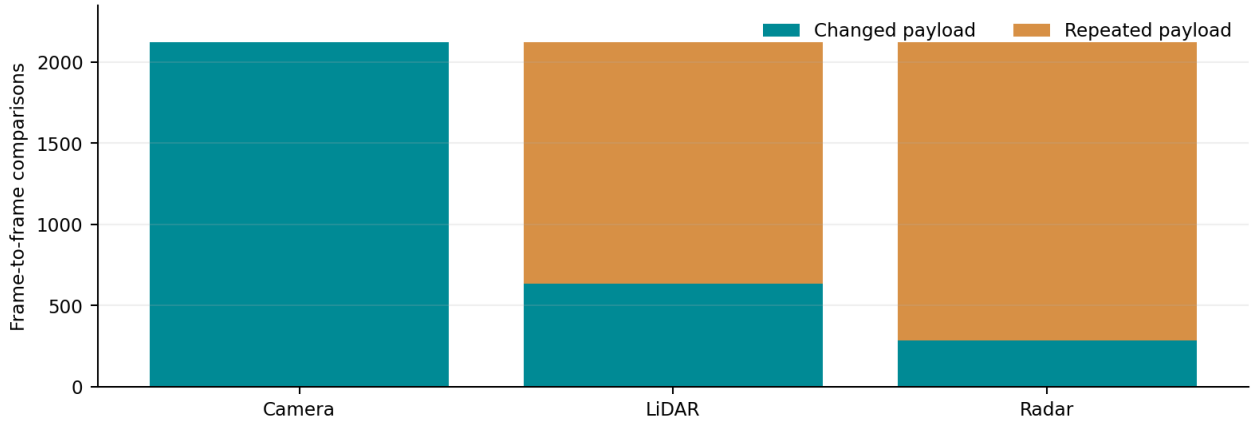
• Separate each sensor’s first observation from changed and repeated payload comparisons.

• Measure unchanged-content intervals and investigate extended LiDAR/Radar repeats at reflection level.

• Cross-check recording/configuration provenance, timestamps and original archived INI SHA-256.

• Keep acquisition age UNKNOWN; repeated content alone must not be classified as a hardware failure.

• Export health tables, repetition runs, reflection audit and a final sensor-health report.



Observed: Camera changed 2,121 times and repeated 0; LiDAR changed 633 and repeated 1,488; Radar changed 286 and repeated 1,835. Each sensor also has one initial observation. Maximum unchanged-content durations: LiDAR 17.669 s, Radar 22.265 s. Effective sensor acquisition rates remain unknown.

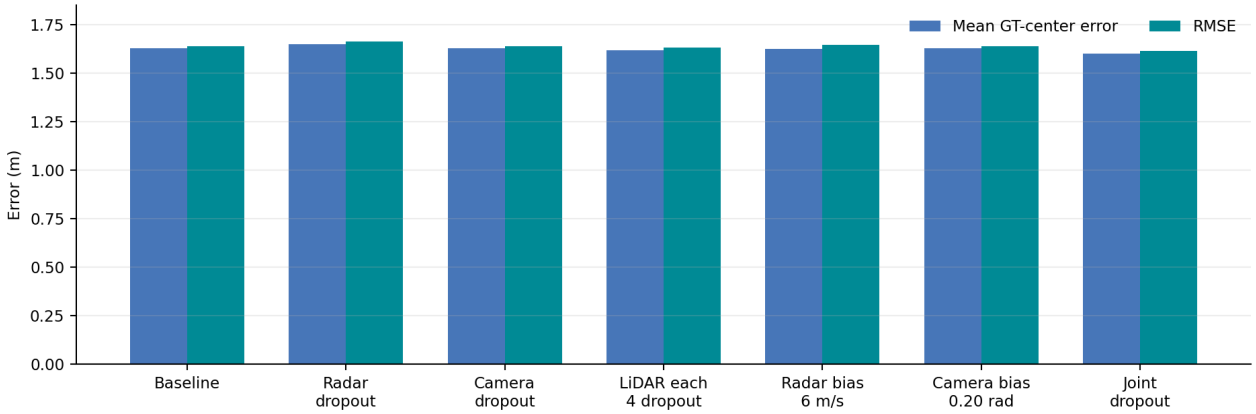
Critical distinction: a new SensorView timestamp or changed Camera payload does not establish independent sensor acquisition time or detection quality.

Evidence: Task25_2_Sensor_Health_Final_Report.md; task25_20_c02_sensor_health_fullrate.csv

Task 25.3 | Fault response and GT-center error

Steps carried out

- Verify matched replay frame identifiers and 23 GT evaluation samples in each scenario.
- Audit injected dropouts and biases against the EKF accepted/rejected update status.
- Compute mean error, RMSE and change from the baseline for the seven scenarios.
- Generate MATLAB fault-error and update-response figures (PNG and PDF).
- Document why lower mean error under dropout is not proof of increased robustness.



Reconstructed from the archived 25.3 summary; each scenario has 23 temporally related GT comparisons.

Scenario	Mean (m)	RMSE (m)	Mean Δ vs baseline (m)
Baseline	1.628213	1.640308	0
Radar dropout	1.650425	1.663593	+0.022212
Camera dropout	1.628205	1.640300	−0.000008
LiDAR periodic dropout	1.618054	1.632332	−0.010159
Radar +6 m/s bias	1.625637	1.645515	−0.002576
Camera +0.20 rad bias	1.628205	1.640300	−0.000008
Joint dropout	1.601929	1.613303	−0.026284

Update behavior: Baseline had four accepted Radar updates and one Camera update. Radar +6 m/s bias retained one Radar update while three were rejected. Camera +0.20 rad bias caused one Camera rejection. Periodic LiDAR dropout suppressed five LiDAR measurements; the joint-dropout condition suppressed four.

Interpretation: no multi-recording significance or generalization is established. The apparent lower errors in some dropout scenarios are not evidence that removing sensors improves perception.

Evidence: Task25_3_Controlled_Fault_Final_Report.md; task25_27 and task25_28 MATLAB plots

Task 25.4 | Association reliability

Steps carried out

• Audit 53 selected frames from the original 2,122-frame C02 V2 recording.

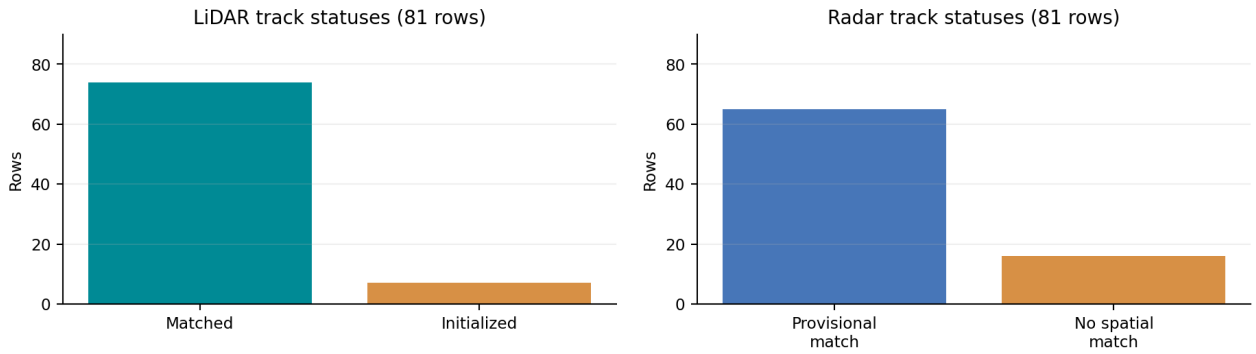
• Separate return-level Radar box-gating from track-level Radar spatial association.

• Count LiDAR initialized and temporally matched track records.

• Compute association distance statistics and generate distribution/timeline visualizations.

• Label Camera bearing geometric compatibility as experimental and uncalibrated.

• Keep identity correctness provisional because independent GT target-identity validation is unavailable.



Population / metric	Observed result
LiDAR temporal track records	81: 74 MATCHED, 7 INITIALIZED
Radar track records	81: 65 PROVISIONAL_SPATIAL_MATCH, 16 without match
Radar return-level candidates	670 PROVISIONAL_BOX_GATE
Camera-bearing support	1 uncalibrated, geometrically compatible bearing
Mean LiDAR temporal distance	0.094688 m (74 valid distances)
Mean Radar box-gap distance	0.044080 m (65 provisional distances)

Do not equate: LiDAR temporal distance and Radar spatial box gap are different geometric measures; neither proves calibrated sensor accuracy or verified target identity. A zero gap to a bounding box is not confirmation that the tracked vehicle is correct.

Evidence: Task25_4_Association_Reliability_Final_Report.md; task25_33_association_distance_analysis.csv

Task 25.5 | EKF statistical consistency and uncertainty

Steps carried out

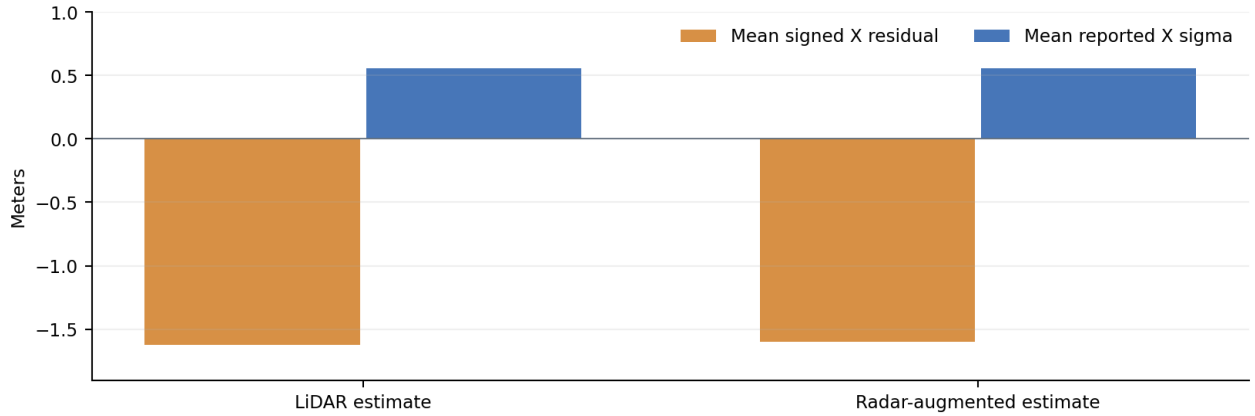
• Collect 24 selected EKF Track 2 observations and 23 GT-center X residual comparisons.

• Identify accepted finite Radar NIS values and distinguish unavailable observations.

• Compare signed X residuals with the estimator’s reported X sigma.

• Export independent consistency audit data and MATLAB Radar NIS, residual/sigma and normalized-residual plots.

• Preserve original data; avoid post-hoc bias correction or unsupported full-state NEES claims.



Ground-truth diagnostic: all 23 paired samples for each estimator exceeded its reported one-sigma magnitude in X. Mean X residuals: LiDAR −1.623522 m; Radar-augmented −1.601283 m. Mean reported sigma: 0.555641 m and 0.552601 m respectively.

Accepted Radar frame	NIS
245	0.591499
286	1.060251
326	0.611538
816	0.019422

These four accepted finite NIS measurements, from a single dependent series, cannot establish statistical calibration. Twenty of the 24 NIS entries are unavailable, not zeros.

Evidence: Task25_5_EKF_Statistical_Consistency_Final_Report.md; task25_40 and task25_43 CSVs

Task 25.6 | Offline BEV geometry validation

Steps carried out

• Inspect the archived C02 V2 source code, sensor configuration and coordinate-extraction conventions.

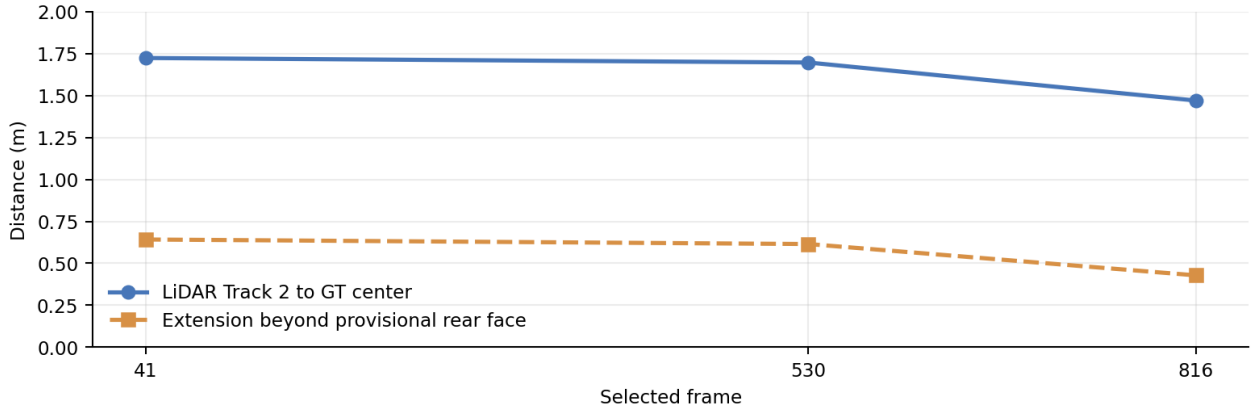
• Verify nominal source translations: LiDAR (2.60, 0, 0.70) m; Radar (2.00, 0, 0.80) m.

• Reconstruct relative GT centers and selected LiDAR/Radar points in ego-referenced bird’s-eye view.

• Audit 59 coordinate rows across frames 41, 530 and 816; compute Track 2 geometric separations.

• Generate a three-frame BEV view and geometry-offset visualization as PNG/PDF.

• Document that nominal mount agreement is source-level consistency only, not independent OSI frame calibration.



Frame	LiDAR Track 2–GT center	Beyond provisional rear face
41	1.725 m	0.642 m
530	1.698 m	0.615 m
816	1.471 m	0.428 m

Interpretation: LiDAR track coordinates are cluster medians; they should not be interpreted as verified vehicle centers. The roughly 1.6 m X discrepancy is consistent with an unresolved representation or coordinate-reference mismatch, but these three frames do not prove its exact source or an appropriate correction.

The study is a BEV diagnostic visualization, not a trained BEV model or certified calibrated fusion system.

Evidence: Task25_6_BEV_Geometric_Validation_Final_Report.md; task25_49 and task25_50 CSVs

Task 25.7 | Reliability-aware fallback policy

Steps carried out (25.7A–G)

• 25.7A: Collect sensor status vocabulary, controlled fault labels and provenance.

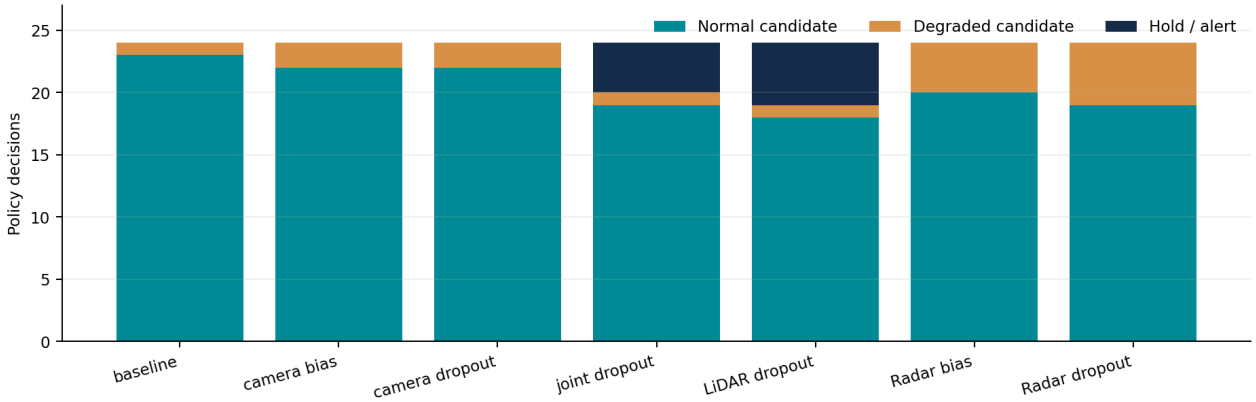
• 25.7B–C: Define 21 machine-readable policy rules and document candidate-mode eligibility.

• 25.7D: Implement the offline fallback controller (Python), producing per-sample decisions and seven scenario summaries.

• 25.7E: Independently audit sensor action assignments, modes, output permissions and authorization flags.

• 25.7F: Recalculate and visualize scenario comparisons and mode timelines using MATLAB.

• 25.7G: Consolidate results, limitations, policy tables and final report.



Policy meaning: NORMAL_CANDIDATE means LiDAR available with no explicitly excluded update; DEGRADED_CANDIDATE means LiDAR available but another sensor update excluded; HOLD_OR_ALERT means LiDAR unavailable and fresh valid-result authorization withheld. All 168 output quality authorizations remain UNVERIFIED.

Decision audit: 504 measurement-action assignments, 168 modes, 168 permissions, 168 UNVERIFIED flags and 21 scenario-summary checks, with zero reported audit discrepancies. This checks internal policy consistency, not externally validated safety.

Evidence: Task25_7_Reliability_Aware_Fallback_Final_Report.md; task25_65, task25_66, task25_67 CSVs

Task 25.8 | Hard perception quality and latency constraints

This stage specified six quality gates that must be justified and independently validated before assigning a perception result the status VALID. The output of Task 25.8 is a prospective validation framework, not six passed quality tests.

Gate	Name	Evidence / unresolved requirement
G01	GT_CENTER_ACCURACY	Historical errors exist; geometry/threshold unresolved
G02	VALID_OUTPUT_COVERAGE	Provisional decisions cannot count as valid outputs
G03	ASSOCIATION_INTEGRITY	Identity validation independent of box gating needed
G04	UNCERTAINTY_CONSISTENCY	Full covariance / appropriate NIS-NEES tests needed
G05	P99_PROCESSING_LATENCY	Processing latency not measured; fixed deadline needed
G06	OVERALL_VALID_RESULT	Fail-closed authorization only if required gates pass

Every Task 25.8 subtask

Subtask	Work completed	Main output / status
25.8A	Inventory available quality evidence and references	7 scenario inventory rows; no validated thresholds
25.8B	Specify the six hard validation gates	6 machine-readable gate records; UNVERIFIED
25.8C	Audit evidence against each gate	161 historical GT evaluation entries, 0 gate passes
25.8D	Define prospective measurement protocol	Required schema, metrics, statistical tests and timing boundaries
25.8E	Build software/hardware readiness matrix	G01–G04 software-first; G05 instrumented CPU then FPGA
25.8F	Produce consolidated quality-constraint report	Task25_8_Hard_Perception_Quality_Constraints_Final_Report.md

Task 25.8C identified 159 provisional outputs and 9 hold/alert decisions, 65 provisional Radar associations, four accepted finite Radar NIS values and 2,122 frames with unknown independent acquisition age. None of these is proof that an accuracy, coverage, uncertainty or p99 latency gate passes.

Quality thresholds were not retrospectively fitted to the already-observed error values. Later measurements need defined denominators, calibration conventions, fixed deadlines and independent validation observations.

Evidence: Task25_8_Hard_Perception_Quality_Constraints_Final_Report.md; task25_71 through task25_76 CSVs

Task 25.9 | Independent Python-MATLAB reproducibility

Every Task 25.9 subtask

Subtask	Verification action	Outcome
25.9A	Inventory 11 CSV datasets, 5 MATLAB scripts, 24 PNG/PDF figures and 5 findings files	3 SHA-256 manifests
25.9B	Calculate selected fallback, association, EKF, NIS and quality metrics independently in Python	18 metrics
25.9C	Recalculate same metrics in MATLAB and compare with strict tolerances	Initially 11 pass / 7 float fail; corrected 18 pass
25.9D	Recheck 11 dataset hashes and preserve both failure evidence and original script	11/11 dataset hashes; 0 residual failures
25.9E	Create final cross-language research report with limitations	18/18 metrics; max abs difference 2.22e-16

Precision discrepancy and documented resolution

The first MATLAB comparison introduced unnecessary numeric-to-string-to-numeric conversions, causing seven floating-point comparisons to fail (largest observed difference about 1.31×10^{-5}). The script was corrected to retain numeric columns as doubles and parse only actual text columns. The accepted numerical agreement tolerances were not relaxed: integer counts exact, floating-point absolute tolerance 1×10^{-10} and relative tolerance 1×10^{-9}.

Final result: all 18 selected metrics agreed; maximum difference $2.220446049250313 \times 10^{-16}$. Original pre-fix script and failed results were preserved for traceability. This is a computational reproducibility result, not independent physical verification.

Example verified metric	Value
Fallback normal / degraded / hold	143 / 16 / 9
LiDAR matches; Radar provisional spatial matches	74; 65
Mean association distances	0.094688 m LiDAR; 0.044080 m Radar gap
Mean signed X residuals	−1.623522 m LiDAR; −1.601283 m Radar
Mean accepted Radar NIS	0.570677 across 4 values
Quality gates authorized	0 of 6

Evidence: Task25_9_Python_MATLAB_Cross_Validation_Final_Report.md; task25_80, task25_81 and task25_83 evidence

Task 25.10 | Research consolidation, preservation and publication

The work was completed through a sequential audit, archive, Git staging and push workflow. These steps belong to documentation and reproducibility, not new perception experiments.

ID	Action	Verified result
A	Inventory folders, reports, filenames, sizes and source SHA-256 hashes	136 pre-manifest research files; 7 folders; 8 stage reports then
B	Recheck original inventory and stage evidence	136/136 files; 8/9 reports; identified missing 25.1 retrospective report
C	Consolidate historical Task 25.1 replay evidence	168 replay rows, 7 summaries, 7 × 24 grouping; 25.1 final report created
D	Validate final source archive readiness	136 original files + 5 documented additions = 141 source files
E	Create ZIP and verify each file in archive	143 entries verified; 2,688,332 bytes
F	Review scientific documentation, archive digest and local Git worktree	9/9 reports, 18/18 metric matches; clean local tree
G	Inspect repository layout, branch, remote and destination	Task 25 destination exists; no tracked files initially
H	Inspect Git ignore rule and nested ZIP; copy/verify source into local repo	143 copied; 142 normal + 1 ignored ZIP; SHA-256 match
I	Scan copied files for sensitive text patterns	0 credential-pattern blockers; 2,080 path hits
J	Classify Windows path findings	2,080 research/tool-path findings across 42 files
K	Inventory non-text files and format signatures	12 PDFs; 15 PNGs; 2 SVGs; 1 ZIP
L	Identify SVGs and PDF metadata field names	SVG signatures; Creator/Producer/Title for 12 PDFs
M	Check SVG active patterns, PNG chunks and limited PDF actions	2 SVG and 12 PDF unflagged; 15 PNG tEXt flagged for review
N	Review PNG text metadata without disclosing values	27 metadata entries: 15 Software, 12 Creation Time; 0 flags
O	Finalize publication scope and limited PDF metadata scan	143 proposed files; 36 PDF fields; 0 flags
P	Stage approved files; force-add only one verified nested ZIP	143 Git objects verified; 0 unrelated changes
Q	Commit after preserving original trailing whitespace in evidence	143 additions; local commit 95dac5a5...
R	Push dedicated research branch; verify GitHub ref	Remote and local SHA equal; clean working tree

Original research ZIP details

Archive path on user workstation: E:\GEMSTAR_Project\sensor-health-monitor\tests\Task_25_Final_Research_Archive.zip. The ZIP was verified entry-by-entry against source hashes. Its SHA-256 is 389EE40F721A0AFCFECE25162A76C43E83FF2716592D7AAE4249B38B1ACB5B7B. The ZIP is a local evidence archive, not proof of perception quality.

GitHub publication details

Branch: research/task25-fusion-reliability. Commit: 95dac5a5eb50f85861a6b6963d1ea85859cc7e46. Exactly 143 additions were confirmed; GitHub returned the same remote commit SHA. One small nested starter ZIP was explicitly force-added because the repository ignored *.zip; it contains task25_replay.py and README.md. The branch was published but not merged into the default branch.

Published Task 25 research folder (GitHub)

Scientific interpretation | What the results support

Supported by recorded evidence	Not yet established
2,122-frame payload-change and repeat audit	Measured acquisition age or proven sensor hardware failure
Seven related matched offline replay scenarios	Statistically generalizable fault robustness
Internal consistency of a conservative fallback policy	Independent valid-output or safety authorization
Provisional spatial and temporal association distances	GroundTruth-verified identity association accuracy
Selected GT residual diagnostics and four finite Radar NIS	Calibrated EKF covariance, valid full-state NEES
Nominal mount translations consistent with archived C02 V2 configuration	Complete OSI / GEMSTAR physical coordinate calibration
18 matched Python/MATLAB summary metrics	New independent physical experiments
Integrity-checked 143-file GitHub archive	p99 latency compliance or measured FPGA power savings

Open technical issues to carry into Task 26 and later work

• Define GT-center and LiDAR measured-point representation consistently; verify frame transforms, orientation and target identity.

• Separate repeated payload content from acquisition age and do not use repeated data as statistically independent measurements.

• Add independent association labels, covariance records and a properly designed NIS/NEES calibration test.

• Record CPU processing-time boundaries with a monotonic clock and measure actual per-result p99 latency against a pre-established deadline.

• Pre-register acceptance metrics and limits before evaluating holdout or fresh matched scenarios.

• Measure system energy on the actual target platform. Evaluate energy per validated perception result and report withheld outputs honestly.

• Keep hardware-accelerated, CPU-only and offline replay comparisons matched on input, configuration, measurement scope and quality requirements.

Completion statement

Task 25 research workflow: COMPLETE. Final-stage reports, source manifests, plots and scripts were preserved and the verified research package published on GitHub. Perception-quality authorization: UNVERIFIED. The work demonstrates an auditable offline research foundation; it does not demonstrate a certified three-sensor fusion system, real-time ARM-FPGA processing, or measured energy savings.

Evidence register and figure locations

The tables and reconstructions in this report were based on the nine published Task 25 final-stage reports and completion audit outputs. Original high-resolution MATLAB figures, CSV datasets, scripts, configuration hashes and audit manifests are in the Task_25_Final_Research repository folder.

Stage	Primary final report	Primary plots / data
25.1	Task25_1_Baseline_Replay_And_Fault_Injection_Final_Report.md	task25_replay_samples.csv; task25_fault_summary.csv
25.2	Task25_2_Sensor_Health_Final_Report.md	task25_20_c02_sensor_health_fullrate.csv; task25_22, _23 SVG
25.3	Task25_3_Controlled_Fault_Final_Report.md	task25_27 and _28 MATLAB PNG/PDF
25.4	Task25_4_Association_Reliability_Final_Report.md	task25_33 CSV; task25_34, _35, _36 PNG/PDF
25.5	Task25_5_EKF_Statistical_Consistency_Final_Report.md	task25_40, _43 CSV; _44, _45, _46 PNG/PDF
25.6	Task25_6_BEV_Geometric_Validation_Final_Report.md	task25_49, _50 CSV; _51, _52 PNG/PDF
25.7	Task25_7_Reliability_Aware_Fallback_Final_Report.md	task25_65, _66, _67 CSV; _68, _69 PNG/PDF
25.8	Task25_8_Hard_Perception_Quality_Constraints_Final_Report.md	task25_71 to _76 CSV
25.9	Task25_9_Python_MATLAB_Cross_Validation_Final_Report.md	task25_77 to _83 audit / numerical evidence

Graphic provenance note: The six chart images in this PDF were reconstructed from numerical values in the published final reports rather than copied from the existing research PNG/PDF files. They are explanatory summaries; the original figures in 04_Plots and the raw numeric CSVs remain the primary research artifacts.

Publication trail: GitHub branch research/task25-fusion-reliability; verified commit 95dac5a5eb50f85861a6b6963d1ea85859cc7e46; all 143 research files checked before publication. The standalone PDF created here is a later summary artifact, so it is not part of that original 143-file commit or immutable ZIP.

Document date: 10 November 2026. Document purpose: concise but comprehensive teaching and research-reference account of each Task 25 stage and its principal subtasks, inputs, outputs, figures, conclusions and unresolved limitations.